

Multi-Agent Release Readiness & Deployment Assistant: Hackathon Submission

1. Problem Statement: The Manual Release Bottleneck

Release management represents the most critical yet vulnerable juncture of the SDLC. Manual oversight at this final gate creates a systemic risk to deployment velocity and production stability, often neutralizing the gains achieved through agile development. For "Elena," the Release Manager, the daily operational bottleneck involves parsing disparate package.json files and performing manual Git diffing across multiple microservices to ensure architectural alignment. This extreme cognitive load and the error-prone nature of drafting release notes under pressure significantly increase Mean Time to Resolution (MTTR) and threaten SDLC parity. There is a strategic requirement for an intelligent, automated assistant to transform this chaotic manual state into a governed, proactive workflow.

2. Solution Statement: Multi-Agent Release Readiness & Deployment Assistant

This solution represents a paradigm shift from reactive scripting to proactive automation, enforcing deployment guardrails and ensuring auditability through automated evidence collection. The architecture functions as a unified bridge between **IBM Bob 2.0**, **watsonx.ai**, and **watsonx Orchestrate**. We utilize **IBM Granite** models within the **watsonx.ai Prompt Lab** to interpret code intent with high fidelity, while **watsonx Orchestrate** manages the overarching business workflows. The implementation prioritizes modular subagent isolation to prevent context pollution—a critical requirement for maintaining governance at scale in complex enterprise environments. By moving beyond simple task assistance to full agentic orchestration, we transform the deployment pipeline from a structural hurdle into a distinct competitive advantage.

3. Technical Execution: IBM Bob 2.0 Utilization

IBM Bob 2.0 serves as the core engine, utilizing specialized "Agent Mode" and "Document Understanding" to parse complex codebase changes autonomously. The architectural implementation leverages specific Bob 2.0 capabilities to maximize interoperability:

- **Agent Mode & Subagents:** Used to isolate focused tasks, such as dependency scanning and commit analysis, within self-contained contexts to prevent context window bloat and ensure architectural integrity.
- **Parallel Tasks:** Implemented to handle high-velocity release cycles by executing multi-step task sequences and repository scans concurrently.
- **Document Understanding:** Targets Elena's specific pain point by automatically analyzing local Git diffs and package.json configurations to identify versioning requirements and potential breaking changes.
- **Custom Rules:** Applied to enforce organizational standards, ensuring absolute consistency in how commits are categorized across the enterprise. The solution results in the automated generation of a comprehensive CHANGELOG.md and a robust audit trail, including the required **task session summary screenshots**. By replacing hours of

manual compilation with seconds of automated reasoning, we reduce integration overhead and ensure that every production push is documented, verified, and ready for deployment.